

Analysis 2: HW1

Aamod Varma

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Problem 3

1. We need to show that the set $\overline{\{x : f * \psi_k(x) \neq 0\}}$ is a compact set. It is enough to show it's closed and bounded. Now note that if $f * \psi_k(x)$ is non-zero then we know it is necessarily true that for some y we have $f(y) \neq 0$ and $\psi_k(x-y) \neq 0$. In other words we need y in support f and $x-y$ in support ψ_k or x in $\text{supp } \psi_k + y$. Now note that we need to consider each $y \in \text{supp } f$. So we need to show that $\text{supp } \psi_k + \text{supp } f = \{x + y \mid x \in \text{supp } \psi_k, y \in \text{supp } f\}$ is compact. Clearly it's bounded because for any x, y as defined in our set if we have $|x| \leq M$ (as support ψ is compact and hence bounded) and $|y| \leq N$ then clearly we have $|x + y| \leq |x| + |y| \leq M + N$ for all x, y and hence the sum is bounded. Now we need to show that it's closed. In other words show that every convergent sequence converges within the set. Consider any sequence (z_n) where z_i is in $\text{supp } \psi_k + \text{supp } f$ now clearly $z_i = x_i + y_i$ where $x_i \in \text{supp } \psi_k$ and $y_i \in \text{supp } f$ by definition. Assume that $(z_n) \rightarrow z$. Now note that we have $(x_n) \rightarrow x$ and $(y_n) \rightarrow y$, we have this because both the supports are compact any sequence has a convergent subsequence. Now this implies $(z_n) = (x_n + y_n) \rightarrow (x + y)$ (note that technically we're considering the convergent subsequence of x and y but because (z_n) is convergent by definition, it is enough to find the limit of any subsequence of (z_n) as all subsequence of a convergent subsequence share the same limit). But x and y are in the supports of ψ_k and f which imply $x + y \in \text{supp } \psi_k + \text{supp } f$ and hence (z_n) converges to some limit that is within the set, making it closed. As we have closed and bounded we have compactness.

Now we need to show smoothness. To show smoothness we essentially need to show that $\partial_x^k(f * \psi_k(x))$ exists for any $k \in \mathbb{N}$. Start with $\partial(f * \psi_k)(x)$ and note that we're essentially doing,

$$\lim_{h \rightarrow 0} \frac{\int_{\mathbb{R}^n} f(y) \psi_k((x + hu) - y) dy - \int_{\mathbb{R}^n} f(y) \psi_k(x - y) dy}{h}$$

This can be rewritten as,

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} f(y) \frac{(\psi_k(x + hu - y) - \psi_k(x - y))}{h} dy$$

We can now use weak dominated convergence to take the limit inside. To apply this first we note that we have $\lim_{h \rightarrow 0} \frac{(\psi_k(x + hu - y) - \psi_k(x - y))}{h} = \partial_x \psi_k(x - y)$ just by definition of derivative - this gives us pointwise convergence. Now note that we also bound $f(y) \frac{(\psi_k(x + hu - y) - \psi_k(x - y))}{h}$ above by some function of $\partial_x \psi_k$ we

get this because ψ_k is bounded - being smooth itself and the supports being compact. This allows us to take the limit inside the integral to get,

$$\partial_x(f * \psi_k)(x) = \int_{R^n} f(y) \partial_x \psi_k(x - y) dy$$

However now note that we can continue doing this because $\partial_x^n \psi_k(x - y)$ exists because ψ_k is a smooth function and hence is infinitely differentiable. Hence for any n we can have $\partial_x^n(f * \psi_k)(x) = \int_{R^n} f(y) \partial_x^n \psi_k(x - y) dy$. This gives us smoothness of $f * \psi_k(x)$

2. We need to show that $\sup |f * \psi_k - f| \rightarrow 0$ as $k \rightarrow \infty$. So we have,

$$|f * \psi_k(x) - f| = \int_{R^n} f(y) \psi_k(x - y) dy - f(x)$$

now we have $\int_{R^n} \psi_k(x) dx = 1$ so we can use this to get,

$$\begin{aligned} |f * \psi_k(x) - f| &= \int_{R^n} f(y) \psi_k(x - y) dy - \int_{R^n} f(x) \psi_k(x - y) dy \\ &= \int_{R^n} (f(y) - f(x)) \psi_k(x - y) dy \end{aligned}$$

Now as f is continuous and has a compact support it is uniformly continuous and for any ε we have a δ such that when $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$. Now we note that using this δ as $k \rightarrow \infty$. Now as ψ has a compact support we can cover it with a ball of radius say R . Now note that as $k \rightarrow \infty$ the support shrinks in size as well and hence the radius R goes to zero. We see this as, if we need $\psi_k(x) > 0$ then we need $k^n \psi(kx) > 0$ or $\psi(kx) > 0$ but this means we're outside the ball we made so kx must be outside the ball and hence we just need a ball of radius R/k i.e a smaller ball, implying as $k \rightarrow \infty$ the ball where ψ_k is non-zero shrinks.

Now note that this means as $k \rightarrow \infty$ we just need to consider the integral over the ball R/k which goes to zero, but as f is uniformly continuous we can ensure that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. So we just need to take k large enough such that ψ_k is zero outside this δ ball. So we have something like,

$$|f * \psi_k(x) - f(x)| \leq \left(\sup_{|x-y| \leq R/k} |f(y) - f(x)| \right) \int_{R^n} \psi_k(x - y) dy$$

note the reason why we just need to consider the supremum inside the ball of radius R/k is because ψ_k is zero outside anyways so the values of $f(y) - f(x)$ don't contribute. Now note that the integral on the right anyways goes to 1 and as f is uniformly continuous for all ε greater than zero we have some δ for which $|f(y) - f(x)| < \varepsilon$ now we just need to pick k such that $R/k < \delta$ or $k > R/\delta$. Which will then give us,

$$\sup |f * \psi_k - f| \leq \varepsilon$$